R Programming – 10 Questions with Code & Solutions

Beginner-friendly R programs, calculations, plots and interpretations.

1. Covariance and Correlation

```
age <- c("5-6","7-8","9-10")
A <- c(18,2,20)
B <- c(22,28,10)
C <- c(20,40,40)

pref <- data.frame(A,B,C)

# 1. Covariance between B and C
cov(B, C)

# 2. Covariance matrix
cov(pref)

# 3. Correlation between B and C
cor(B, C)

# 4. Correlation matrix
cor(pref)
```

Explanation: cov() measures how two variables vary together. cor() gives the standardized relationship from -1 to +1.

Result: Using the three age-group rows as observations, cov(B,C) = -72. Correlation(B,C) ≈ -0.866. The full matrices are produced by cov(pref) and cor(pref). The study is problematic because age is grouped and the data are aggregated as frequencies; covariance/correlation of B and C does not itself test the association between age and photograph preference. A chi-square test of independence is more appropriate for the 3×3 frequency table. There may also be confounding from how children were sampled and from treating categorical preferences as numerical variables.

2. Equal-frequency partitioning, bin means, bin boundaries and histogram

```
x <- c(18,18,18,20,20,20,20,20,20,20,21,21,21,21,
        25,25,25,25,25,28,28,30,30,30)

# Equal-frequency bins: 3 bins
bins <- split(x, rep(1:3, each = length(x)/3))
bins

# Bin means
mean_bins <- lapply(bins, mean)
mean_bins

# Smoothing by bin means
smooth_mean <- unlist(lapply(bins, function(b) rep(mean(b), length(b))))
smooth_mean

# Smoothing by bin boundaries
smooth_boundary <- unlist(lapply(bins, function(b) {
  lo <- min(b); hi <- max(b)
  sapply(b, function(v) if (v - lo <= hi - v) lo else hi)
}))
smooth_boundary

# Histogram
hist(x, breaks = 3,
     main = "Histogram after Equal-Frequency Division",
     xlab = "Values", col = "lightblue")
```

Explanation: There are 24 values, so 3 equal-frequency bins contain 8 values each. The bins are formed from the sorted data. Bin-mean smoothing replaces every value in a bin with that bin's mean; boundary smoothing replaces each value with the nearest bin boundary.

Result: Bin 1 = 18,18,18,20,20,20,20,20; Bin 2 = 20,20,21,21,21,21,25,25; Bin 3 = 25,25,25,28,28,30,30,30. The histogram visualizes the frequency distribution.

3. Compare Class A and Class B

```
A <- c(76,35,47,64,95,66,89,36,84)
B <- c(51,56,84,60,59,70,63,66,50)

mean(A); median(A); range(A)
mean(B); median(B); range(B)

boxplot(A, B,
        names = c("Class A","Class B"),
        main = "Year 9 Exam Scores",
        ylab = "Marks")
```

Explanation: mean() gives average, median() gives the middle value, and range() gives minimum and maximum values.

Result: Class A: mean ≈ 65.78, median = 66, range = 35–95 (range width 60). Class B: mean ≈ 62.11, median = 60, range = 50–84 (range width 34). Class A has the higher mean and median. Class A also has the wider spread. The boxplot makes the difference in central tendency and variability visible.

4. Min-max and z-score normalization

```
x <- c(200,300,400,600,1000)

# (a) Min-max normalization to [0,1]
min_max <- (x - min(x)) / (max(x) - min(x))
min_max

# (b) Z-score normalization
z_score <- (x - mean(x)) / sd(x)
z_score

# Example from $50,000-$100,000, v=$80,000
(80000 - 50000) / (100000 - 50000)
```

Explanation: Min-max normalization transforms values to a fixed range, here 0 to 1. Z-score normalization subtracts the mean and divides by the sample standard deviation.

Result: Min-max results: 0, 0.125, 0.25, 0.5, 1. The example $80,000 becomes 0.6. For z-score, R's sd() uses sample standard deviation; the resulting values are approximately -0.742, -0.494, -0.247, 0.247, 1.236.

5. Histogram for AirPassengers

```
data("AirPassengers")

hist(AirPassengers,
     breaks = seq(100, 700, by = 150),
     main = "AirPassengers Histogram",
     xlab = "Number of Passengers",
     ylab = "Frequency",
     xlim = c(100, 700))
```

Explanation: The breaks are 100, 250, 400, 550 and 700, giving bins of width 150. xlim starts the displayed x-axis at 100.

Result: Because the AirPassengers data extend beyond 700, observations above the displayed range are not visible in the requested plot. If all observations must be represented, increase the final break/x-axis limit.

6. Multiple Lines in One Plot – mtcars mpg and qsec

```
data(mtcars)

plot(mtcars$mpg, type = "o", col = 1,
     ylim = range(c(mtcars$mpg, mtcars$qsec)),
     xlab = "Car Index",
     ylab = "Value",
     main = "MPG and QSEC")

lines(mtcars$qsec, type = "o", col = 2)

legend("topright",
       legend = c("mpg","qsec"),
```

```
        col = c(1,2), lty = 1, pch = 1)
```

Explanation: plot() creates the first line and lines() adds the second line to the same graph. mpg and qsec have different scales, so this is mainly a visual comparison. For a statistically meaningful comparison, standardization or separate axes may be preferable.

7. Water dataset: mortality vs hardness and linear regression

```
# If package MASS is not installed:
# install.packages("MASS")

library(MASS)
data("water")

plot(water$hardness, water$mortality,
     main = "Mortality vs Water Hardness",
     xlab = "Hardness",
     ylab = "Mortality",
     pch = 19)

model <- lm(mortality ~ hardness, data = water)
abline(model)

summary(model)

# Prediction at hardness = 88
predict(model, newdata = data.frame(hardness = 88))
```

Explanation: The scatterplot checks whether mortality and hardness appear linearly related. lm() fits a straight-line regression model, and predict() estimates mortality for hardness 88.

Result: The regression output gives the slope, intercept, R-squared and p-value. The exact predicted value is generated by R from the installed MASS water dataset.

8. Boxplot of mpg by number of cylinders

```
data(mtcars)

boxplot(mpg ~ cyl, data = mtcars,
        main = "MPG by Number of Cylinders",
        xlab = "Number of Cylinders",
        ylab = "Miles per Gallon",
        col = "lightblue")
```

Explanation: The formula mpg ~ cyl creates one boxplot for each cylinder group. The box shows the median and interquartile range, whiskers show the typical range, and isolated points can indicate potential outliers.

Result: In general, cars with fewer cylinders tend to have higher mpg, while cars with more cylinders tend to have lower mpg.

9. Tennis players: detecting scoring outliers with a boxplot

```
points <- c(12,15,17,18,20,21,22,23,24,25,26,27,28,45)

boxplot(points,
        main = "Tennis Player Points",
        ylab = "Points Scored",
        col = "lightgreen")

# Identify boxplot outliers
boxplot.stats(points)$out
```

Explanation: First collect the points scored by each player. A boxplot displays the median, quartiles and whiskers. Points beyond the whisker limits are flagged as potential outliers.

Result: The value 45 is expected to appear as a potential outlier because it is much higher than the main group. An outlier should be investigated rather than automatically deleted; it may represent exceptional performance or a data-entry error.

10. BloodPressure vs Age using diabetes.csv

```
# Put diabetes.csv in the working directory
data <- read.csv("diabetes.csv")

# Check column names
names(data)

# Scatterplot
plot(data$Age, data$BloodPressure,
     main = "Blood Pressure vs Age",
     xlab = "Age",
     ylab = "Blood Pressure",
     pch = 19)

# Add regression line
model <- lm(BloodPressure ~ Age, data = data)
abline(model)

# Create age groups
data$AgeGroup <- cut(data$Age,
                     breaks = c(0,20,30,40,50,60,100),
                     right = FALSE)

# Mean blood pressure by age group
bp_mean <- aggregate(BloodPressure ~ AgeGroup,
                     data = data, FUN = mean)
bp_mean

# Bar chart
barplot(bp_mean$BloodPressure,
        names.arg = bp_mean$AgeGroup,
        main = "Average Blood Pressure by Age Group",
        xlab = "Age Group",
        ylab = "Average Blood Pressure",
        las = 2,
        col = "orange")
```

Explanation: The scatterplot shows the relationship between age and blood pressure. The bar chart compares average blood pressure across age groups. The regression line helps visualize the overall linear trend.

Result: The exact age group with the highest average blood pressure depends on the contents of the supplied diabetes.csv. The code calculates it directly instead of assuming a result.

Important correction for Question 1

The table contains frequencies of photograph choices within age groups. Therefore, covariance/correlation between the columns B and C is not the correct inferential test for whether age and photograph preference are significantly related. A chi-square test of independence should be used on the original 3×3 frequency table. In R:

```
tab <- matrix(c(18,22,20,2,28,40,20,10,40), nrow=3, byrow=TRUE)
chisq.test(tab)
```

This directly tests whether photograph preference is independent of age group.